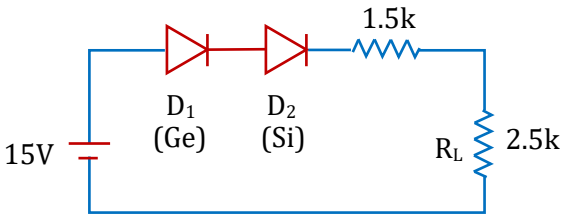
RACE
Regular Analysis Through Continuous Exercise

Semiconductors : RACE-01

1. In the given circuit, the voltage across load resistance (R_L) is:

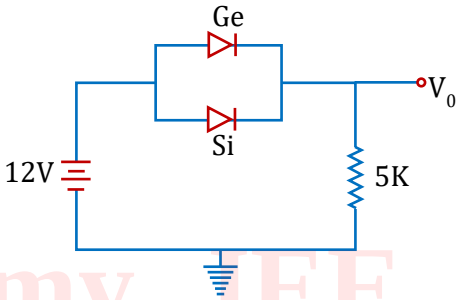


(A) 8.75 V (B) 9.00 V (C) 8.50 V (D) 14.00 V

PSD135

2. Ge and Si diodes start conducting at 0.3 V and 0.7 V respectively. In the following figure if Ge diode connection are reversed, the value of V_o changes by:

(Assume that the Ge diode has large breakdown voltage)



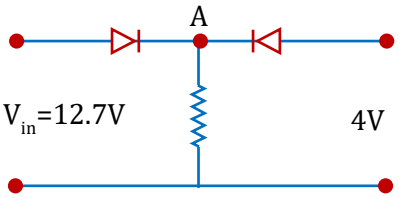
(A) 0.6 V (B) 0.8 V

(C) 0.4 V (D) 0.2 V

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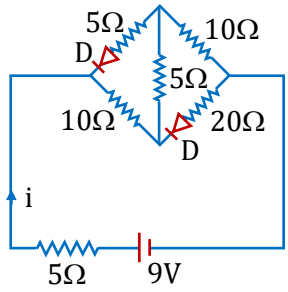
PSD136

3. Both the diodes used in the circuit shown are assumed to be ideal and have negligible resistance when these are forward biased. Built in potential in each diode is 0.7 V. For the input voltages shown in the figure, the voltage (in Volts) at point A is _____ .



PSD137

4. The current i in the network is :

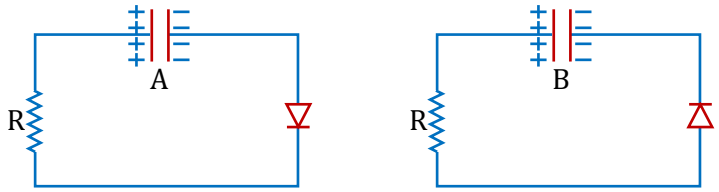


(A) 0 A (B) 0.6 A (C) 0.3 A (D) 0.2 A

PSD138

For More Join: @IITJEE_Advanced

5. Two identical capacitors A and B, charged to the same potential $5V$ are connected in two different circuits as shown below at time $t = 0$. If the charge on capacitors A and B at time $t = CR$ is Q_A and Q_B respectively, then (Here e is the base of natural logarithm)



(A) $Q_A = VC, Q_B = \frac{VC}{e}$

(B) $Q_A = \frac{CV}{2}, Q_B = \frac{VC}{e}$

(C) $Q_A = VC, Q_B = CV$

(D) $Q_A = \frac{VC}{e}, Q_B = \frac{CV}{2}$

PSD139

6. With increasing biasing voltage of a photodiode, the photocurrent magnitude

(A) increases initially and saturates finally.

(B) increases initially and after attaining certain value, it decreases.

(C) increases linearly.

(D) remains constant.

PSD140

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7. For extrinsic semiconductors; when doping level is increased:

(A) Fermi-level of p-type semiconductor will go upward and Fermi-level of n-type semiconductors will go downward.

(B) Fermi-level of p-type semiconductors will go downward and Fermi-level of n-type semiconductor will go upward.

(C) Fermi-level of both p-type and n-type semiconductors will go upward for $T > T_F\ K$ and downward for $T < T_F K$, where T_F is Fermi temperature.

(D) Fermi-level of p and n-type semiconductors will not be affected.

PSD141

8. The photodiode is used to detect the optical signals. These diodes are preferably operated in reverse biased mode because:

(A) Fractional change in majority carriers produce higher forward bias current.

(B) Fractional change in majority carriers produce higher reverse bias current.

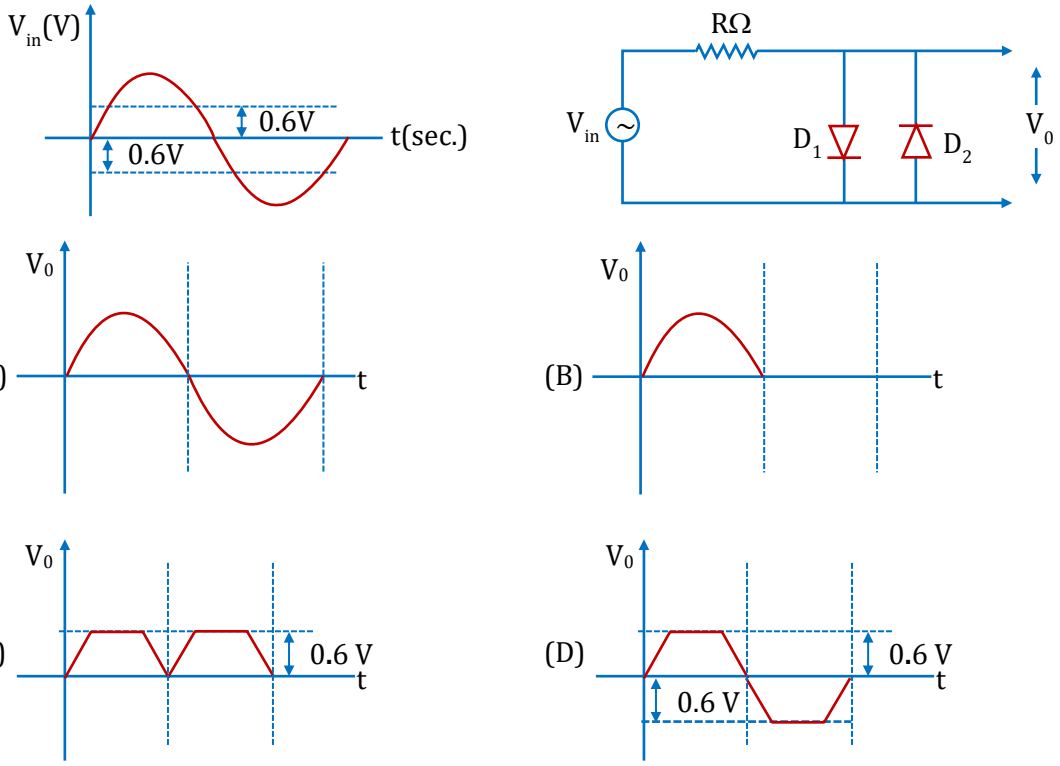
(C) Fractional change in minority carriers produce higher forward bias current.

(D) Fractional change in minority carriers produce higher reverse bias current.

PSD142

For More Join: @IITJEE_Advanced

9. In the given circuit the input voltage V_{in} is shown in figure. The cut–in voltage of p–n junction diode (D_1 or D_2) is 0.6 V. Which of the following output voltage (V_0) waveform across the diode is correct?



PSD143

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10. Match the List-I with List-II

	List-I		List-II
A.	Intrinsic Semiconductor	I.	Fermi-level near conduction band
B.	n-type semiconductor	II.	Fermi-level at middle
C.	p-type semiconductor	III.	Fermi-level near valence band
D.	Metals	IV.	Fermi-level inside conduction band

Choose the **correct** answer from the options given below:

(A) (A) → I, (B) → II, (C) → III, (D) → IV

(B) (A) → II, (B) → I, (C) → III, (D) → IV

(C) (A) → II, (B) → III, (C) → I, (D) → IV

(D) (A) → III, (B) → I, (C) → II, (D) → IV

PSD144

ANSWER KEY

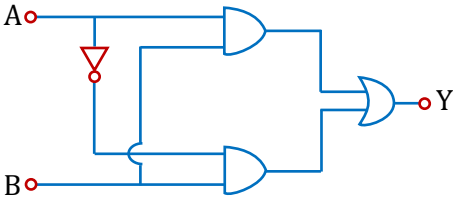
Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	12	C	A	A	B	D	D	C

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Semiconductors : RACE-02

1. The truth table for this given circuit is :



(A)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(C)

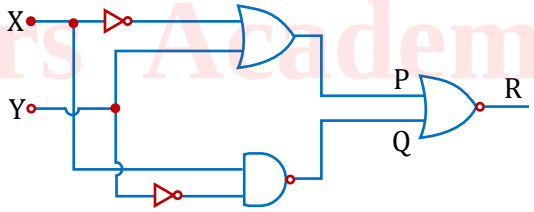
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(D)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

PSD145

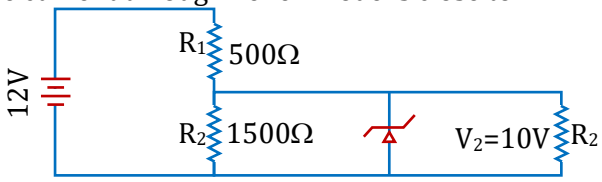
2. To get output '1' at R, for the given logic gate circuit the input values must be :



(A) $X = 0, Y = 1$ (B) $X = 1, Y = 1$ (C) $X = 0, Y = 0$ (D) $X = 1, Y = 0$

PSD146

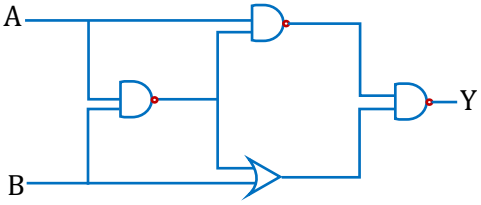
3. In the given circuit the current through Zener Diode is close to :



(A) 6.0 mA (B) 4.0 mA (C) 6.7 mA (D) 0.0 mA

PSD147

4. The output of the given logic circuit is :

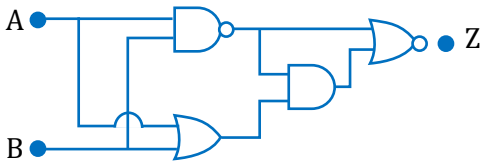


(A) $\bar{A}B$ (B) $A\bar{B}$ (C) $AB + \overline{AB}$ (D) $A\bar{B} + \bar{A}B$

PSD148

For More Join: @IITJEE_Advanced

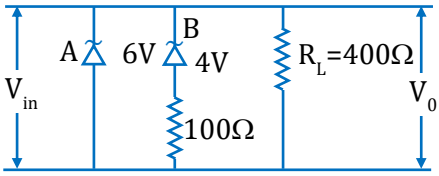
5. In the following digital circuit, what will be the output at 'Z', when the input (A, B) are (1,0), (0,0), (1,1), (0,1):

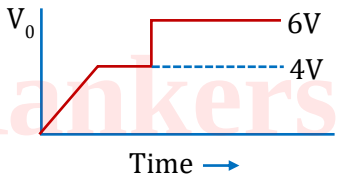


(A) 1, 0, 1, 1 (B) 0, 1, 0, 0 (C) 0, 0, 1, 0 (D) 1, 1, 0, 1

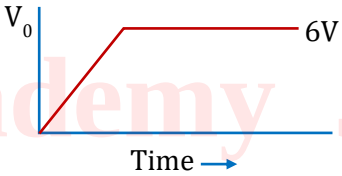
PSD149

6. Two Zener diodes (A and B) having breakdown voltages of $6V$ and $4V$ respectively, are connected as shown in the circuit below. The output voltage V_0 variation with input voltage linearly increasing with time, is given by: ($V_{input} = 0$, V at $t = 0$) (Figures are qualitative)

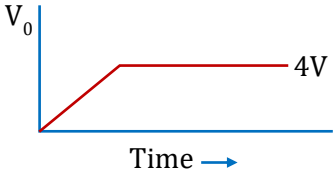




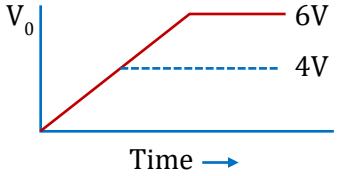
(A)



(B)



(C)

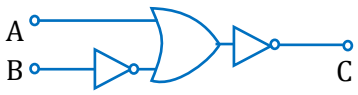


(D)

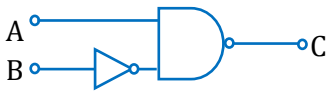
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PSD150

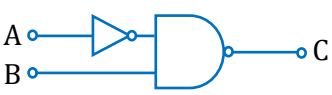
7. The logic circuit shown above is equivalent to :



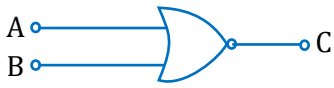
(A)



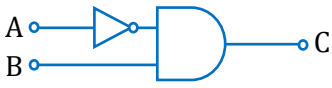
(B)



(C)



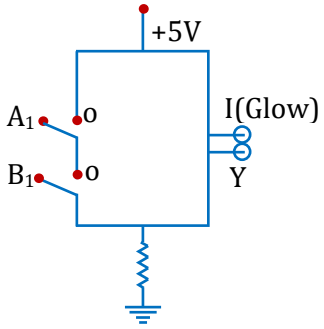
(D)



PSD152

For More Join: @IITJEE_Advanced

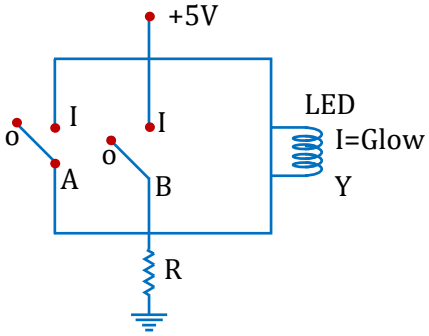
8. The logic gate equivalent to the given circuit diagram is :



(A) OR (B) NAND (C) NOR (D) AND

PSD153

9. Name the logic gate equivalent to the diagram attached



(A) OR (B) NOR (C) NAND (D) AND

PSD154

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	B	D	D	B	C	D	D	B	B

For More Join: @IITJEE_Advanced